

HEMANTH RAJ M V

Bengaluru, Karnataka

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Education

BMS College of Engineering, Bengaluru

Bachelor of Engineering in Computer Science (Data Science)

Oct 2023 – July 2027 (Expected)

CGPA: 8.32/10

Technical Skills

Languages: Java, Python, SQL, Kotlin, C

ML & Data: Pandas, NumPy, Scikit-learn, LightGBM, Feature Engineering, NLP (NLTK), EDA

Backend & AI: FastAPI, REST APIs, RAG Pipelines, Embeddings, SHAP

Databases & Tools: Firebase, ChromaDB, Redis, SQLite, Git, Docker,

Projects

• RiskGuard — Automated Return-Risk Decisioning Platform

Razorpay AI Buildathon 2026

FastAPI, Next.js/React, Tailwind CSS, LightGBM, SQLite, SHAP

- Built a full-stack **internal decision-support tool** — **FastAPI** backend + **Next.js/React** frontend — that scores e-commerce return risk in real time, **automating a manual fraud-review process**.
- Designed a **chronological 65/15/20 train/validation/test split** to prevent the model from being tuned on future data, and aligned **SHAP** explanations with calibrated risk scores so each decision carries a clear, trustworthy reason.
- Achieved **0.706 AUC (95% CI 0.680–0.732)**, within 0.009 of the estimated performance ceiling (0.715) for the dataset, at a threshold balancing 0.449 precision and 0.470 recall.

• Tether — Social Accountability Android App

GitHub — APK (v1.0)

Kotlin, MVVM, Firebase Auth/Firestore/FCM, Navigation Component

- Built an Android app using **Kotlin/MVVM**; optimized navigation with manual **NavController** handling to eliminate UI flickering and achieve **sub-second tab switching**.
- Engineered a **Foreground Service** for multi-mode focus tracking (Stopwatch/Pomodoro) with **1-second polling accuracy**, keeping sessions alive in the background across app minimization and screen-off states
- Implemented a **real-time nudge system** using **FCM + Firestore** with **sub-second sync**; optimized backend via **composite indexing** to efficiently handle thousands of log entries by group and date.

• Visual Document Q&A — End-to-End RAG Pipeline

GitHub

FastAPI, ChromaDB, Ollama (phi3), PyMuPDF

- Built a **Visual Document Q&A system** with a **FastAPI + ChromaDB** RAG pipeline using `all-MiniLM-L6-v2` embeddings and **Ollama (phi3)**, achieving **~12 tok/s** streaming via SSE with **1s retrieval latency**.
- Engineered a **4-signal Concept Importance Engine** (TF-IDF + spread + heading weight + **PageRank centrality**) and visualized concept relationships as an **Obsidian-style D3.js force-directed graph** with tiered node coloring, co-occurrence edges, and click-to-query interaction.
- Implemented **tiered heading detection** (H1/H2/H3 via PyMuPDF font-size analysis) and **async LLM topic refinement** that renders the TF-IDF graph instantly then merges domain concepts in background — boosting validated terms by **1.5×** with silent fallback ensuring zero UX disruption.

Achievements / Leadership

Nutanix Certified Associate (NCA)

2026

- Validated core competency in **hyperconverged infrastructure (HCI)** and **cloud infrastructure fundamentals**.

GenAI Hackathon — Special Mention

2025

- Recognized for **Visual Document Q&A**, a RAG-based system with FastAPI, ChromaDB, and a D3.js visualization layer.

Treasurer, DSynC Club — BMSCE

April 2025 – Present

- Managed end-to-end club finances across **10+ events**, owning budgeting, expense tracking, and reimbursements. Secured corporate sponsorships for **UTSAV** (60K+ footfall) and **PHASESHIFT** (40K+ footfall)